



Figure 3: Device measurements in Admin UI

```
'X-Sitewhere-Tenant': 'sitewhere1234567890'
```

For a listing of all available sites:

```
GET /sitewhere/api/sites
```

For a listing of a particular site:

```
GET /sitewhere/api/sites/{siteToken}
```

For a listing of the associated devices' locations on a site:

```
GET /sitewhere/api/sites/{siteToken}/locations
```

For a listing of site measurements:

```
GET /sitewhere/api/sites/{siteToken}/measurements
```

For a listing of alerts on a particular site:

```
GET /sitewhere/api/sites/{siteToken}/alerts
```

For a listing of command invocations:

```
GET /sitewhere/api/sites/{siteToken}/responses
```

For a listing of command responses:

```
GET /sitewhere/api/sites/{siteToken}/responses
```

To obtain server runtime information:

```
GET /sitewhere/api/system/runtime
```

To obtain the version of SiteWhere:

```
GET /sitewhere/api/system/version
```

You can explore the Swagger UI or [documentation](http://documentation.sitewhere.org/rest/single.html). [sitewhere.org/rest/single.html](http://documentation.sitewhere.org/rest/single.html) for a complete listing.

NodeRED support

An add-on for NodeRED is available with the name *node-red-contrib-sitewhere*. It provides a few nodes that generate JSON payloads for device registration, location updates, measurements, alerts as well as acknowledgement



Figure 4: The Swagger interface for REST API

events, and can be wired to a regular MQTT publish node. Similarly, a MQTT subscriber node can be wired with a command node for parsing. If you are using Raspberry Pi, you can wire a DHT node or a SenseHAT node with a measurement node for the desired sensor values. Raspberry Pi's integration tutorial mentioned earlier gives insights into using NodeRED to generate the above payloads and parsing commands. You can find detailed examples written in Node.js and NodeRED flows for the communication through MQTT and REST, in the Sitewhere-examples sub-directory of github.com/rajeshsola/iot-examples.

Java agent

A Java agent is available under github.com/sitewhere/sitewhere-tools, which provides a command line client for custom communication with the platform. You can edit the *config/config.properties* and run the pre-built JAR file for initial testing. You can also edit the example code for custom communication or integrate this agent with any Java code base easily.

Mobile apps

An Android SDK is available under github.com/sitewhere/sitewhere-android-sdk, which provides *SiteWhereActivity* and *SiteWhereProtobufActivity* for base functionality and communication services like MQTT and REST for client implementation. Android apps can be easily built using this SDK. You may refer to the sample application before proceeding, which sends location events and accelerometer readings as measurements.

Acknowledgement: Figure 1 is taken from <http://documentation.sitewhere.org/architecture.html>.

References

- 1) sitewhere.org/architecture.html
- 2) github.com/sitewhere

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